

Minimum Drawdown and Catastrophic Elevations for Lake Powell

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Overview

This is an R Markdown document. The document generates a plot showing an example Minimum Drawdown and Catastrophic Elevations for Lake Powell.

Minimum Drawdown Elevation: 3525 feet.

Catastrophic Elevation: 3514 feet.

(*Set these elevations on lines 38 to 42 in the .Rmd file).

The plot also includes:

1. A trace showing Lake Powell storage over time going back to May 2024.
2. A text box listing active storage and elevation on the date before the script was run. Also includes the storage above the 2019 DCP protection elevation.
3. Horizontal lines showing the Minimum Drawdown Elevation and the Catastrophic Elevation with buffer volume.

This script also generates a Table showing the Time to react before drawing down to catastrophe. This is calculated using the Minimum Drawdown volume, catastrophe volume, buffer volume, and annual decrease in flow.

This work generates Figure 2 and Table 1 in: Brittany Fager, David Rosenberg, Erik Porse (2026). “Why communicating minimum drawdown elevations for Lake Powell and Lake Mead can mitigate risks of volatile Colorado River flow.” *Utah State University Digital Commons*. <https://doi.org/10.15142/h11w-a657>.

Data wrangling strategy

1. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation’s HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html
2. Import the reservoir elevation-volume curves and critical elevations.
3. Filter the reservoir data for monthly Lake Powell storage.
4. Filter the daily reservoir data for the current Lake Powell and Lake Mead storage/elevation

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), “Minimum Drawdown and Catastrophic Elevations for Lake Powell.” Utah State University. Logan, Utah. <https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/MinimumDrawdownElevations>.

Current Storage and Elevations

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## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/MinimumDrawdownElev
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
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## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "Reservoir Data on: June 25 , 2026"
```

ResName	Storage	Pool Elevation
Lake Powell	5.64	3526.72
Lake Mead	7.34	1045.19

Figure 2. Lake Powell Minimum Drawdown and Catastrophic Elevations.

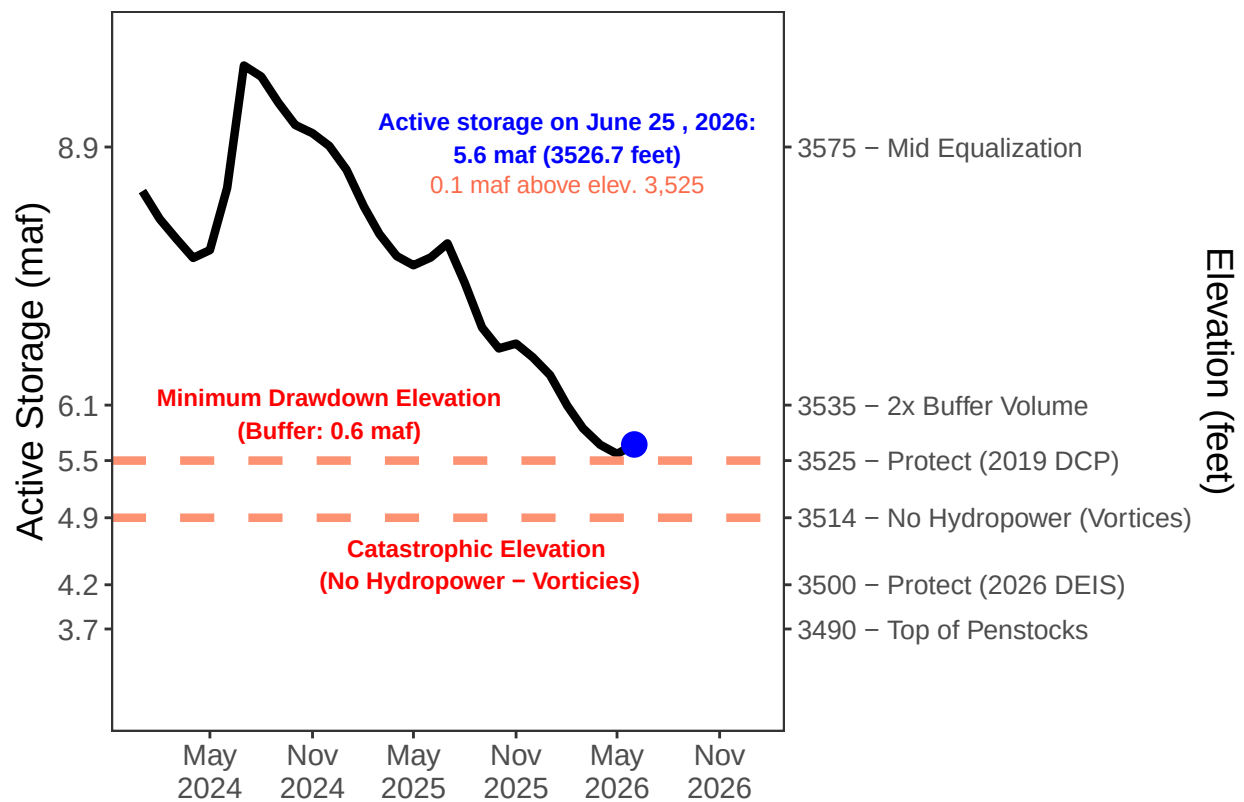


Table 1. Time to react.

##	Attribute	Value	Round	Units
## 1	a. Minimum Drawdown Elevation (3525 feet)	5.5		maf
## 2	b. Catastrophic Elevation (3514 feet)	4.9		maf
## 3	c. Buffer [a - b]	0.6		maf
## 4	d. Annual decrease in flow	2.0		maf per year
## 5	e. Time to react [c / d]	0.3		years